Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

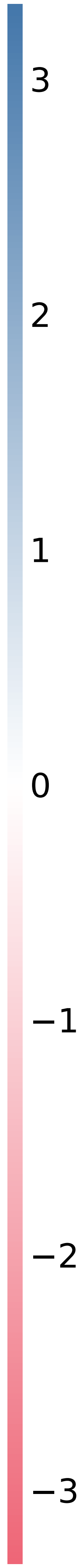
F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV



Reasoning + Answer:

<think>

Use the classical electron energyâĢĵmomentum relation for relativistic de Broglie wavelength:

Î» = h/(âĪļ(2mE) c),

where h is PlanckâĢĻs constant (in ergÂ·s), m is the electron mass (9.11 ÃĹ10^-28 g), c is the speed of light (3.00 ÃĹ10^10 cm/s), and E = 1 eV = 1.60 ÃĹ10^-12 erg.

Calculate the factor:

âĪļ(2mE/c^2) = âĪļ[(2 ÃĹ9.11 ÃĹ10^-28 ÃĹ1.60 ÃĹ10^-12)/(3.00 ÃĹ10^10)^2] âīĪ1.875 ÃĹ10^-23 gÂ·cm/s.

Then Î» = h/(âĪļ(2mE) c) = (6.625 ÃĹ10^-27 ergÂ·s)/(1.875 ÃĹ10^-23 gÂ·cm/s ÃĹ3.00 ÃĹ10^10 cm/s) âīĪ1.23 ÃĹ10^-7 cm.

Correct answer: G. 1.23 ÃĹ10^-7 cm

</think>

<answer>

G. 1.23 ÃĹ10^-7 cm

</answer>